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MILITARY SPECIFICATION

FLOODLIGHT SET, TRAILER-MOUNTED

This specification is apporved for use within the USA Belvoir Reserarch, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trailer-mounted floodlight set complete with a telescoping tower, three outdoor floodlights, each with a 1000-watt metal halide lamp, three outdoor remote ballasts for reliable operation of the floodlights, a 12-gallon fuel tank, four outriggers/leveling jacks, a splash panel, and a control box assembly, all mounted on a two-wheeled, pneumatic-tired, military standard trailer. The floodlight set has provisions for accepting electrical power from an external source, such as a nearby commercial power hook up or a separate mobile power unit. In additon, an open space with mounting provisions is provided on the trailer for mounting a Department of Defense 3 kW, 60 Hz diesel or gasoline engine-driven generator set. However, a generator set is not provided with the floodlight set. The telescoping tower consists of four sections which permit the tower to be extended to any height from approximately 12 to 40 feet above the ground. The tower is mounted on the floodlight set in a horizontal (transport) position. Using an integral winch, operating personnel can raise the tower from the horizontal position to an upright position and then extend it to the desired operating height. The floodlights are stored on the trailer but for operation they are mounted on top of the tower and produce a large, circular-type light pattern. The outriggers/leveling jacks are used to provide stability of the floodlight set when the tower is extended. For protection against mud thrown up by the wheels of the towing vehicle, a splash panel is provided. The 12-gallon fuel tank, in combination with any fuel tank supplied with a generator set, provides for extended operation of the floodlight set. Independent control and circuit protection for the three floodlights is provided by three circuit breakers in the control box assembly. A ground rod and hammer are also provided.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.

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MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-M-8090	- Mobility, Towed Aerospace Ground Equipment, General Requirements for.
MIL-T-10579	- Trailer, Cargo, 2 Wheel, 1/4 Ton to 1-1/2 Ton.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MS52152	- Chassis, Trailer: 3/4 Ton, 2 Wheel, Military Design, M116A2.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13226E4350 - Floodlight Set, Trailer Mounted.

PUBLICATIONS

FORMS

DA Form 2258 - Depreservation Guide for Vehicles and Equipment.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commericial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this

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specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The trailer-mounted floodlight set (hereinafter called "floodlight set") shall be in accordance with top assembly drawing TA13226E4350 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances are permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the floodlight set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations or production floodlight sets from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish floodlight sets conforming to this specification.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contactor and shall be subject to the provisions of 3.3.1 thru 3.3.4 of this specification.

3.3.1 Material deterioration prevention and control. The floodlight set shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or teated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the floodlight set may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin

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raw materials. The components, pieces and parts incorporated in the floodlight set may be newly fabricated from recovered materials to the maximum extent practicable, provided the floodlight set produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the floodlight set.

3.4 Safety.

3.4.1 Nonfunctional sharp edges. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.5 Design and operating requirements. The floodlight set design and operating requirements shall conform to the following.

3.5.1 Weight. The gross weight of the floodlight set with the 3 kW generator set, including all gasoline and oil, shall not exceed 2780 pounds.

3.5.1.1 Center of gravity. With the floodlight set in the towing configuration (see 6.4.4) and with a 3 kW generaot set mounted on the trailer, the load on the trailer lunette shall not exceed 200 pounds. The center of gravity of the floodlight set utilizing the 3 kW generator set shall be identified on all floodlight sets in accordance with TA13226E4350.

3.5.2 Twelve-hour operation. When tested as specified in 4.5.2.2, the flood-foodlight set shall operate at full load (see 6.4.3), without damage (see 6.4.1), for not less than 12 hours, 8 hours of which shall be continuous, and shall not be damaged as a result of perfoming the operating and shutdown procedures specified in 6.4.2.1 and 6.4.2.2.

3.5.3 Environmental conditions.

3.5.3.1 Operational conditions. When tested as specified in 4.5.2.3, the floodlight set shall be capable of operating (see 6.4.2) at full load without damage under the following conditions:

a. Temperatures from -20 to 110 °F.
b. Relative humidity up to 100 percent.
c. Rainfall.
d. Shock (rail impact).

3.5.3.2 Storage and transit conditions. When tested as specified in 4.5.2.3, the floodlight set shall not be damaged by storage in temperatures from -30 to +145 °F for relative humidity up to 100 percent.

3.5.4 Towing mobility. When tested as specified in 4.5.2.4, the floodlight set shall meet the requirements of MIL-M-8090 for type III, group A mobility without damage.

3.5.5 Tower strength, floodlight set stability. When tested as specified in 4.5.2.5, the floodlight set, with the tower and outriggers/leveling jacks extended, shall withstand, without damage, 50 mph winds, and shall not overturn.

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3.5.6 Reliability, availability, and maintainability (RAM).

3.5.6.1 Reliability. When tested as specified in 4.5.2.6, the floodlight set shall demonstrate a reliability of not less than 97.5 percent for a 13-hour mission. A mission shall consist of 20-30 minutes erection time (including lamp warm-up), 12 hours operation time, and 20-30 minutes post-operation maintenance and disassembly of equipment.

3.5.6.2 Maintainability.

3.5.6.2.1 Scheduled maintenance. When tested as specified in 4.5.2.7, the floodlight set, excluding the generator set and trailer, shall demonstrate a minimum of 250 hours of operation between scheduled maintenance actions. Scheduled maintenance actions shall take not more than 2 man-hours to perform.

3.5.6.2.2 Maintenance ratio. The floodlight set, excluding the generator set and trailer, shall have a maintenance ratio of not more than 0.02. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hour for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenace schedule shall be furnished prior to the start of any testing.

3.6 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish in accordance with the requirements of MIL-V-173, composition I or II, as applicable (see 3.6.1), with 7 ± percent salicylanilide or 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetrically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and parepared so that the surfaces are free from all foreign matter which would interfere with the adherence of function of the varnish.

3.6.1 Composition. Compositon II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.7 Identification marking. The floodlight set shall be identified in accordance with TA13226E4350.

3.8 Instruction plate. The instruction (operation) plate shall be in accordance with TA13226E4350.

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3.9 Schematic and wiring diagrams. Schematic and wiring diagram plates shall be in accordance with TA13226E4350.

3.10 Treatment and painting. Treatment and painting of the floodlight set shall conform to MIL-T-704.

3.10.1 Externally-exposed surfaces. Unless otherwise specified herein, all externally exposed surfaces of the floodlight set shall be cleaned, treated, and painted in accordance with MIL-T-704, type F, with the top coat conforming to MIL-C-46168, color green 383. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704. Samples of paints made to the green 383 requirement shall be forwarded to US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-VO, Fort Belvoir, VA 22060-5606 for approval prior to acceptance. A minimum of a quart sample from each production lot shall be forwarded.

3.10.2 Unpainted surfaces. The sliding portions of the tower assembly and the glass lens of the floodlight hoursing shall be cleaned and treated, but shall not be painted. In addition, fuel lines and power cables (including the flood-light cable) shall not be painted.

3.11 Government-furnished property. The following property in the quantitites indicated will be furnished by the Government (see 6.5):

Description	Identification	Quantity for each floodlight set
Chassis, Trailer, M116A2, 3/4 Ton, 2 Wheel	NSN 2330-01-101-8434 MS52152	1

3.12 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantitities indicated will be loaned by the Government (see 6.6):

Description	Identification	Quantity for each first article floodlight set
Military Standard Mobile (skid) Engine-generator set, 3 kW, 60 Hz	See 6.6	1

3.13 Workmanship.

3.13.1 Castings and forgings. All parts, components, and assemblies of the floodlight set which include castings and forgings shall be clean of harmful

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extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.13.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by nethods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or when they are not detrimental to safety.

3.13.3 Welding. Welding shall be in accordance with TA13226E4350.

3.13.4 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

3.13.5 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washer or lockwashers shall be provided where necessary. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasterners shall be correctly torqued and shall have full thread engagement. Bolts shall protude nor more than 2 full threads.

3.13.6 Riveted connections. Rivets shall fill the holes completely. The upset heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the Contract, or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inpsections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article shall be tested as specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for rejection. When specified (see 6.2) a first article test report shall be provided.

4.4. Quality conformance inspection.

4.4.1 Quality conformance sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Quality conformance examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Quality conformance tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 4.0 percent defective.

4.5. Inspection procedure.

4.5.1 Examination. The floodlight set shall be examined as specified in 4.3.1 (first article examination) and 4.4.2 (quality conformance examination for the defects listed in table I.

TABLE I. Examination schedule (requirements).

Number	Defects	Requirement Paragraph
Major		
101.	Floodlight set not in accordance with TA13226E4350.	3.1
102.	Material not as specified.	3.3
103.	Material deterioration prevention and control not as specified.	3.3.1
104.	Use dissimilar metals not as specified.	3.3.2

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TABLE I. Examination schedule (requirements). Cont'd).

Number	Defects	Requirement Paragraph
Major		
105.	Identification of materials and finishes not as specified.	3.3.3
106.	Nonfunctional sharp edges, projecting points, and excessive length of fastening devices not avoided as specified.	3.4.1
107.	Weight not as specified.	3.5.1
108.	Load on the trailer lunette not as specified and center of gravity not identified on all floodlight sets as specified.	3.5.1.1
109.	Fungus and moisture resistance not as specified.	3.6
110.	Instruction plate not as specified.	3.8
111.	Schematic and wiring diagrams not as specified.	3.9
112.	Castings and forgings not as specified.	3.13.1
113.	Metal fabrication not as specified.	3.13.2
114.	Welding not as specified.	3.13.3
115.	Brazing not as specified.	3.13.4
116.	Bolted connections not as specified.	3.13.5
117.	Riveted connections not as specified.	3.13.6
118.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the floodlight sets.	3.3.4
Minor		
201.	Identification marking not as specified.	3.7
202.	Treatment and painting not as specified.	3.10
203.	Externally exposed surfaces not cleaned, treated and painted in accordance with MIL-T-704, type F as specified.	3.10.1
204.	Unpainted surfaces not cleaned and treated as specified.	3.10.2

4.5.2 Tests.

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4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter, at the climatic conditions existing at the place of test, with a Government-loaned 3 kW engine-generator set installed on floodlight set but using commercial external power to operate the floodlight set, and with the floodlight set in the towing configuration. Any test failure attributable to Government-furnished or Government-loaned equipment shall not constitute failure of any test specified herein.

4.5.2.2 Twelve-hour operating test. The Government-loaned 3 kW engine-generator set shall be used to supply power to the floodlight set for this test. Perform the operating procedures specified in 6.4.2.1, "a" through "g", and operate the floodlight set at full load (see 6.4.3) for not less than 12 hours, 8 hours of which shall be continuous. After 12 hours of operation, perform the shutdown procedures specified in 6.4.2.2, "a" through "f". Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3 Environmental tests. The floodlight set shall be subjected to the following tests of MIL-STD-810:

a. High temperature, method 501.2, procedure I (storage) and procedure II (operation).

For procedure I, the storage cycle shall consist of:

(1) maintaining the internal chamber temperature at 91 °F for 3 hours
(2) lowering the temperature to 86 °F within 1 hour and maintaining at 86 °F for 2 hours;
(3) raising the temperature to 145 °F within 7 hours and maintaining at 145 °F for 4 hours; and
(4) lowering the temperature to 91 °F within 7 hours.

The storage cycle shall be repeated for a total of seven 24-hour cycles. During each 24-hour cycle, the relative humidity shall be maintained: (1) between 33 and 41 percent at 91 °F; (2) between 41 and 44 percent at 86 °F; and (3) between 5 and 6 percent at 145 °F.

For procedure II, the operation cycle shall consist of:

(1) maintaining the internal chamber temperature at 91 °F for 3 hours;
(2) lowering the temperature to 86 °F within 1 hour and maintaining at 86 °F for 3 hours;
(3) raising the temperature to 110 °F within 6 hours and maintaining at 110 °F for 5 hours; and
(4) lowering temperature to 91 °F within 6 hours.

The operation cycle shall be repeated for a total of three 24-hour cycles. During each 24-hour cylce, the relative humidity shall be maintained: (1) between 33 and 41 percent at 91 °F; (2) between 41 and 44 percent at 86 °F; and (3) between 14 and 16 percent at 110 °F. The floodlight set shall be operated at full load during the time the temperature is at 110 °F.

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b. Low temperature, method 502.2, procedure I (storage) and procedure II (opration). For procedure I, the storage temperature shall be -30 °F and it shall be maintained for 24 hours. For procedure II, the operating temperature shall be -20 °F.

c. Humidity, method 507.2, procedure I (natural) and procedure II (induced). For procedure I, cycle 3 (cyclic high humidity) shall be conducted using the 24-hour cycle 3 curve of figure 507.2-1. For procedure II, cycle 5 (cyclic high humidity) shall be conducted using the 24-hour cycle 5 curve of figure 507.2-2. Cycles 3 and 5 shall be repeated as indicated in table 507.2-II for non-hazardous items.

d Rain, method 506.2, procedure III (watertightness).

e. Shock, method 516.3, procedure VIII (rail impact). The floodlight set shall be impacted twice in each direction of each orientation at a speed of 8 miles per hour for a total of four impacts. The floodlight set shall be packaged in accordance with level A preservation and level A packing of 5.2.1 and 5.3.1, respectively.

The floodlight set shall be installed in the test facility in the towing configuration but with the floodlights removed from the storage area, attached to the tower, and connected for operation. When in the test chamber, operation of the floodlight set shall consist only of operation the floodlights. When operation of the floodlight set is required outside the test chamber, operation shall be as specified in 6.4.2. Nonconformance to 3.5.3 shall constitute failure of these tests.

4.5.2.4 Road mobility. Unless otherwise specified (see 6.2), road tests shall be conducted by the Government in accordance with MIL-M-8090 for group A type III mobility, and with at least 6 gallons of gasoline in the 12-gallon fuel tank. An electrical/operational check-out of the floodlight set shall be conducted in accordance with 6.4.2 on completion of each test course. Nonconformance to 3.5.4 or leakage of gasoline from the 12-gallon fuel tank shall constitute failure of this test.

4.5.2.5 Telescoping tower and outrigger/leveling jack test. With the floodlight set free standing on a smooth, unpaved area having a slope of approximately five degrees, the outriggers extended, and the floodlight set leveled, the tower (with floodlights attached), shall be raised to its maximum height. Readjust the jacks as necessary to keep the floodlight set level. While the tower is fully extended, a static horizontal force, of magnitude to simulate a 50 mph wind, shall be applied to the tower to determine conformance to rigidity construction, and stability requirements. The force shall be directed in one of the four cardinal points and distributed to five points spaced equidistant over the height of the tower. The force shall consist of the force due to the floodlight surface area (force A) and the force due to the wind load over the height of the tower (force B). Force A shall be applied at the top of the tower and force B shall be evenly distributed to the other four points on the tower. For example, with a 30-foot tower and for an example force of 300 pounds directed north, a horizontal component force of 160 pounds (force A) would be applied at the top of the tower and a horizontal component force of 35 pounds would be applied at 6, 12, 18, and 24 feet simultaneouly; all five component forces directed north. Force B in this example, would be 140 pounds. The force shall be determined by the contractor

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for the type and height of the tower, and the type of floodlights used. The floodlight set shall be tested in each direction of each orientation for a total of sixteen tests. With the front of the floodlight set facing directly uphill on the five-degree slope, the force shall be applied and directed successively to the tower in each of the four cardinal points (North, East, South, and West). The force shall be gradually applied and maintained for five minutes at the required value. After testing in each cardinal direction, the tower shall be lowered and raised. Repeat the tests with the front of the floodlight set facing directly downhill on the five-degree slope, and facing directly sideways. Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.6 Reliability. The floodlight set shall be tested in accordance with test plan IV of MIL-STD-781, using mean missions between failure instead of mean-time-between-failure (MTBF). A mission shall be as defined in 3.5.6.1 and consist of the procedures listed in 6.4.2. For test purposes, the specified mean mission between failure is 40 and minimum mean missions between failure is 20. When a failure occurs, that mission shall be terminated, repairs made using standard parts, and the next mission started. Normal preventive maintenance shall be allowed during test. Nonconformance to 3.5.6.1 shall constitute failure of this test.

4.5.2.7 Maintainability.

4.5.2.7.1 Maintenance evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.5.6.2.1. Failure of the floodlight set to operate a minimum of 250 hours between scheduled maintenance actions shall constitute failure of this test.

4.6 Inspection comparison. The Government may select floodlight sets at any time during the production period and subject these floodlight sets to the examination specified in 4.5.1 and tests specified in 4.5.2.1 and 4.5.2.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on floodlight sets selected at random from those which have been accepted by the Government and will not include the previously inspected first article floodlight set. In addition to any test specified as apart of the inspection comparision, the Government reserves the right to conduct any and all other tests contained in this specification as apart of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison floodlight set to meet any requirements specified hferein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison floodlight set and shall be cause for refusal by the Government to continue acceptance of production floodlight sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contactor at no cost to the Government on floodlight sets previously accepted and produced under contract. Any deficiencies found as a result of the inspection comparison will be considered prima facia evidence that

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all floodlight sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be for the cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Quality conformance pack sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Quality conformance pack examination. Samples selected in accordance with 4.7.2.2 shall be examined for the defects listed in table II. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE II. Examination schedule (packaging).

Number	Defects	Requirement Paragraph
Major		
119.	Disassembly for level A or B preservation not as specified.	5.2.1.1, 5.2.2
120.	Matchmarking for level A or B preservation not as specified.	5.2.1.2, 5.2.2
121.	Unprotected surfaces not preserved for level A or B as specified.	5.2.1.3,
122.	Floodlights and tower not preserved for level A or B as specified.	5.2.1.4, 5.2.2
123.	Junction box and control box assembly not preserved for level A or B as specified.	5.2.1.5, 5.2.2
124.	Trailer chassis not preserved for level A or B as specified.	5.2.1.6, 5.2.2
125.	Other components not specifically mentioned not preserved for level A or B as specified.	5.2.1.10, 5.2.2
126.	Fuel tank not preserved for level A or B as specified.	5.2.1.12, 5.2.2
127.	Floodlight set not preserved for commercial in accordance with ASTM D 3951.	5.2.3

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TABLE II. Examination schedule (packaging). (Cont'd)

Number	Defects	Requirement Paragraph
128.	Floodlight set not packed for level A as specified.	5.3.1
129.	The floodlight set not packed for commercial in accordance with ASTM D 3951.	5.3.2
130.	Marking for military level of protection not in accordance with MIL-STD-129.	5.4.1
131.	Marking for commercial packaging not in accordance with ASTM D 3951.	5.4.2
Minor		
205.	Repair parts not preserved for level A or B as specified.	5.2.1.7, 5.2.2
206.	Maintenance tools not preserved for level A or B as specified.	5.2.1.8, 5.2.2
207.	Technical publications not preserved for level A or B as specified.	5.2.1.9, 5.2.2
208.	Consolidation of repair parts, maintenance tools, technical publications, and removed components not preserved for level A or B as specified.	5.2.1.11, 5.2.2
209.	Depreservation guide not protected and furnished as specified.	5.5

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.8). The first article pack may be accomplished utilizing either the first article floodlight set or a production floodlight set. If the first article floodlight set is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, when requested by the Government, to facilitate comparison between the first article floodlight set and production floodlight set.

5.2 Preservation. Degree of preservation shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A preservation.

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5.2.1.1 Disassembly. The floodlights and tower shall be removed. Any additional disassembly shall be the minimum necessary to safeguard parts subject to damage and loss or protruding parts that tend to increase cubage, or as needed to accomplish the preservation and packing specified herein, in accordance with the contractor's recommendation. Bolts, nuts, pins, and washers involved shall be reinstalled in mating parts and secured to prevent loss.

5.2.1.2 Matchmarking. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking information shall be applied with waterproof ink on tags conforming to UU-T-81, type A.

5.2.1.3 Unprotected surfaces. Unpainted exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

5.2.1.4 Floodlights and tower. Floodlights and tower shall be indiviually containerized in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The contents shall be blocked and braced as required to prevent movement or damage.

5.2.1.5 Junction box and control box assembly. Openings into the junction box and control box assembly and exposed ends of the electrical cable shall be sealed with tape conforming to MIL-T-22085, type II. The power cable to the generator set shall be coiled and the coil secured with tape specified herein. The two individual insulated wires shall be tagged to indicate their position for reconnection.

5.2.1.6 Trailer chassis. The trailer chassis components shall be preserved in accordance with the level A requirements of MIL-STD-281.

5.2.1.7 Repair parts. The preservation application criteria and application submethods of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.8 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.2.1.9 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. When specified (see 6.2), technical manuals shall be provided.

5.2.1.10 Other components. Other components of the floodlight set not specifically mentioned shall be preserved as specified for items of similar characteristics and design or as specified for repair parts, and consolidated as specified in 5.2.1.11.

5.2.1.11 Consolidation. Repair parts, maintenance tools, technical publications and removed components not otherwise containerized shall be consolidated in a close-fitting box conforming to MIL-T-22085, type II, overseas type, style optional. Blocking, bracing, or cushioning as required to prevent movement or damage of the contents shall be provided.

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5.2.1.12 Fuel tank. All openings into the fuel tank and fuel lines shall be selaed with tape in accordance with PPP-T-60, type IV.

5.2.2 Level B preservation. The floodlight set shall be preserved the same as level A, except that the trailer chassis components shall be preserved in accordance with the level B requirements of MIL-STD-281 and boxes shall be domestic type.

5.2.3 Commerical preservation. The floodlight set shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Degree of packing shall be level A or commercial, as specified (see 6.2).

5.3.1 Level A packing. The trailer-mounted floodlight set shall be shipped mobile. The boxed floodlights and consolidated containers shall be secured on the trailer, as space permits, in a manner that will provide minimum cube and will not interfere with lifting or towing. The boxed tower and any boxed floodlights or consolidated containers that will not fit on the trailer shall be shipped separate from but at the same time as the trailer-mounted floodlight set. Boxes secured to the trailer shall not require strapping. Boxes not secured to the trailer shall be closed and strapped in accordance with the appendix to the box specification except that strapping shall conform to QQ-S-781, class I, type I or IV, size as applicable; and unless otherwise specified, shall be finish B. When specified (see 6.2), finish A strapping shall be used.

5.3.2 Commercial packing. The floodlight set shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall be in accordance with ASTM D 3951. In addition, exterior containers shall be marked with the appropriate weight and cube.

5.5 Depreservation guide. Two depreservation guides shall be furnished. One copy of a depreservation guide shall be prepared and placed in a waterproof envelope marked "Depreservation Guide" and shall be secured in a conspicuous location on the floodlight set. The other copy shall be placed in the box with the technical publications. Unless otherwise specified (see 6.2), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used (see 6.9).

6. NOTES

6.1 Intended use. The trailer-mounted floodlight set is intended to be used to provide adequate area and point lighting for construction, maintenance, and other service and service support activities as required whenever necessary.

6.2 Ordering data.

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MATERIAL PROCURED UNDER A

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is not required (see 3.2).
c. When Government-loaned property is not required (see 3.12).
d. Time frame required for submission of the first article pack (see 5.1).
e. Degree of preservation and packing required (see 5.2 and 5.3).
f. When finish A strapping is required (see 5.3.1).
g. When other than DA Form 2258 is to be used (see 5.5).
h. When a first article report shall be provided (see 4.3.2).
i. When technical manuals shall be provided (see 5.2.1.9).

6.3 First article inspection. When a first article inspection is required, the floodlight set should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Damage. Damage is deformation, loosening, breakage, change of fit of any component or part, corrosion, or any condition resulting in malfunction of the floodlight set or any component part.

6.4.2 Floodlight set operation. Operation of the floodlight set shall include the following procedures.

6.4.2.1 Operating procedures.

a. Extend outrigger/jack assemblies and level floodlight set.
b. Remove floodlights from storage area and attach to tower.
c. Turn power supply ON.
d. Turn floodlight switches and warning beacon (marker light) switch on control box assembly to ON, allowing 5-second intervals on each floodlight.
e. Operate winch to pivot tower from horizontal (transport) position to vertical (operating) position.
f. Operate winch to tower to maximum height.
g. Readjust jacks as necessary to level floodlight set.

6.4.2.2 Shutdown procedures.

a. Set floodlight switches and warning beacon (marker light) switch on control box assembly to OFF.
b. Turn power supply OFF.
c. Retract the tower to the minimum vetical (operating) height.
d. Lower the tower to the horizontal (transport) position.
e. Remove the floodlights and store in storage area.
f. Retract the outrigger/jack assemblies and secure.

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6.4.3 Floodlight set maximum operating load. The floodlight set maximum operating load shall consist of the three floodlights, lamps, and ballasts, and the required wiring/electrical power cable assemblies.

6.4.4 Floodlight set towing configuration. The floodlight set towing configuration shall be as follows:

a. Outrigger/jack assemblies retracted and secured.
b. Floodlights stored and secured in storage area.
c. Tower lowered to the horizontal position and secured.
d. All switches set OFF position.
e. Nose wheel retracted.
f. Floodlight mounting beam stored and secured.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.11.

6.6 Government-loaned property. The contracting officer should arrange to furnish the property specified in 3.12. The mobile generator set should be diesel or gasoline powered depending on which generator set is required and available.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 First article pack. Any changes or deviations of production packs for approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the floodlight set in accordance with the specification.

6.9 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 5.5).

6.10 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary, including any special tools, and instructions regarding delivery of floodlight sets.

6.11 Changes, deviations, and waivers. All changes, deviations, and waivers to the floodlight set covered by this revision of the specification will be required to meet the performance requirements specified in previous editions.

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6.12 Subject term (key word) listing.

Area lighting
Floodlight set
Lighting
Trailer mounted floodlight set

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - GS

Project 6210-A640

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-52920D(ME)	Floodlight Set, Trailer-Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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